



Molybdate-based conversion treatment for improving the peeling strength between aluminum foil and polypropylene grafted with glycidyl methacrylate

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ABSTRACT

This paper describes an environment-friendly treatment for aluminum (Al) foil that can greatly improve the adhesion strength between foil with the polymer film to a similar extent as the traditional chromate–phosphate treatment. Al foil was treated with different methods, after which a modified polypropylene (PP) film (PP grafted with glycidyl methacrylate, PP-g-GMA) was directly joined with the treated Al foil by hot pressing. The T-peeling strengths of the Al/PP-g-GMA composites were measured, and the peeling strength of molybdate-treated foil was found to be approximately 10 times greater than that of untreated foil. The surface morphologies of the Al foils were observed by scanning electron microscopy and atomic force microscope, their chemical compositions were characterized by energy-dispersive spectroscopy. The hydrophilicities and surface energies of the treated Al foils were further evaluated by contact-angle measurements.

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1. Introduction

Metal–polymer composites are widely used in aerospace, automobile, and packaging industries [1]. Among such composites, Al/polymer (Al/P) composites are attracting considerable research interest because of their desirable properties, such as low weight and excellent impermeability to moisture and air [2]. However, the low adhesive strengths of Al/P composites limit their applications. To overcome this problem, researchers have proposed various methods, such as polymer modification, metal treatment, and adhesive use [3–10].

Surface pretreatment of Al can effectively improve the adhesion and corrosion resistance of Al/P composites. This treatment can be divided into two main categories, namely, physical and chemical treatments. Physical treatments such as mechanical grinding and grit blasting are used to remove loose contaminated layers and to provide a rough surface for mechanical interlocking with an adhesive [11]. Chemical treatments such as chemical conversion treatment and electrochemical treatment can modify the surface by depositing a new coating layer designed to improve coupling with a given adhesive system [12–14]. Compared with physical methods, chemical conversion treatments are more effective and concise for achieving effective adhesion between metal and polymer.

To obtain a strong and stable bond between Al and polymer layers such as polypropylene (PP), the naturally formed surface oxide on Al must be removed and replaced with a new oxide layer that is continuous, solid, and corrosion resistant. The most successful and widely used treatments are chromate conversion coatings [15–19]. However, hexavalent chromate Cr (VI) is highly toxic, and its use has been limited by several European Union Environmental Legislation and Restriction of Hazardous Substances directives since 2006. Therefore, environment-friendly conversion coatings must be developed to replace traditional chromate treatments.

Low toxicity conversion coatings formed in non-Cr (VI) solution such as titanium, zirconium, Cr (III), and cerium salt baths are widely studied [20–23]. Molybdate-based conversion coatings are promising alternatives because of their low toxicity and good corrosion resistance properties similar to those of chromate-based coatings [24–29]. Yong et al. [24] analysed the corrosion protection of a molybdate/phosphate composite conversion coating on magnesium alloy. Karpakam et al. [26] prepared a polyaniline–molybdate coating on steel and investigated its performance as a corrosion barrier.

Some studies have reported on the treatment of Al foil with molybdate-based conversion solution [30–32]. In the present study, an Al foil was treated with a molybdate-based conversion solution, and a composite film of the treated foil with PP film grafted with glycidyl methacrylate (PP-g-GMA) was prepared by hot pressing. Results showed that molybdate-based conversion coatings can effectively improve the T-peeling strength of Al foil with PP-g-GMA film. To explore the underlying mechanism, the surface morphology of Al foil was

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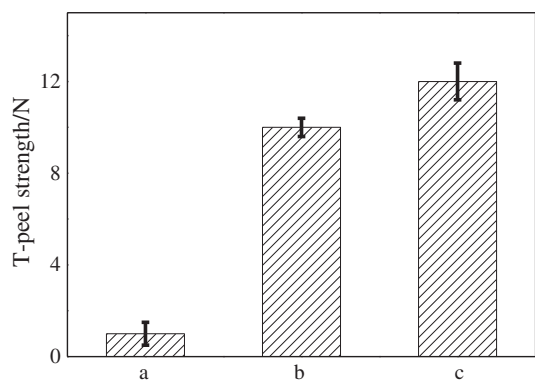


Fig. 1. T-peel strengths of Al foil/PP-g-GMA composites with different pretreatments: (a) no treatment, (b) molybdate-based conversion, and (c) chromate-phosphate conversion.

observed by scanning electron microscopy (SEM) and atomic force microscope (AFM), the chemical composition was characterized by energy-dispersive spectroscopy (EDS). The hydrophilicity and surface energy of the coated Al foil were further evaluated based on contact-angle measurements.

2. Experimental

2.1. Materials

The thickness of the Al foil AA 8021 used in this study was 40 μm . Ammonium molybdate, sodium fluoride, chromium trioxide and phosphoric acid were purchased from Shanghai Ling Feng Chemical Reagent Co., Ltd., China. PP-g-GMA was home-made in our laboratory, and its melt flow index (MI) was approximately 5 g/10 min. PP-g-GMA film with a thickness of 100 μm was prepared by mould pressing.

2.2. Surface pretreatment of Al

2.2.1. Molybdate-based chemical treatment

The substrates were first degreased at 40 $^{\circ}\text{C}$ in acetone for 10 min and subsequently immersed in an aqueous solution containing phosphoric acid (60 g/L), ammonium molybdate (6 g/L), sodium fluoride (2 g/L), and FN-44 (4 g/L, self-developed additive) at 40 $^{\circ}\text{C}$ for 1 min. Finally, the substrates were rinsed with tap water and dried in an oven for 1 h.

2.2.2. Chromate-phosphate conversion treatment

The substrates were first degreased at 40 $^{\circ}\text{C}$ in acetone for 10 min and subsequently immersed in a solution containing hexavalent chromium

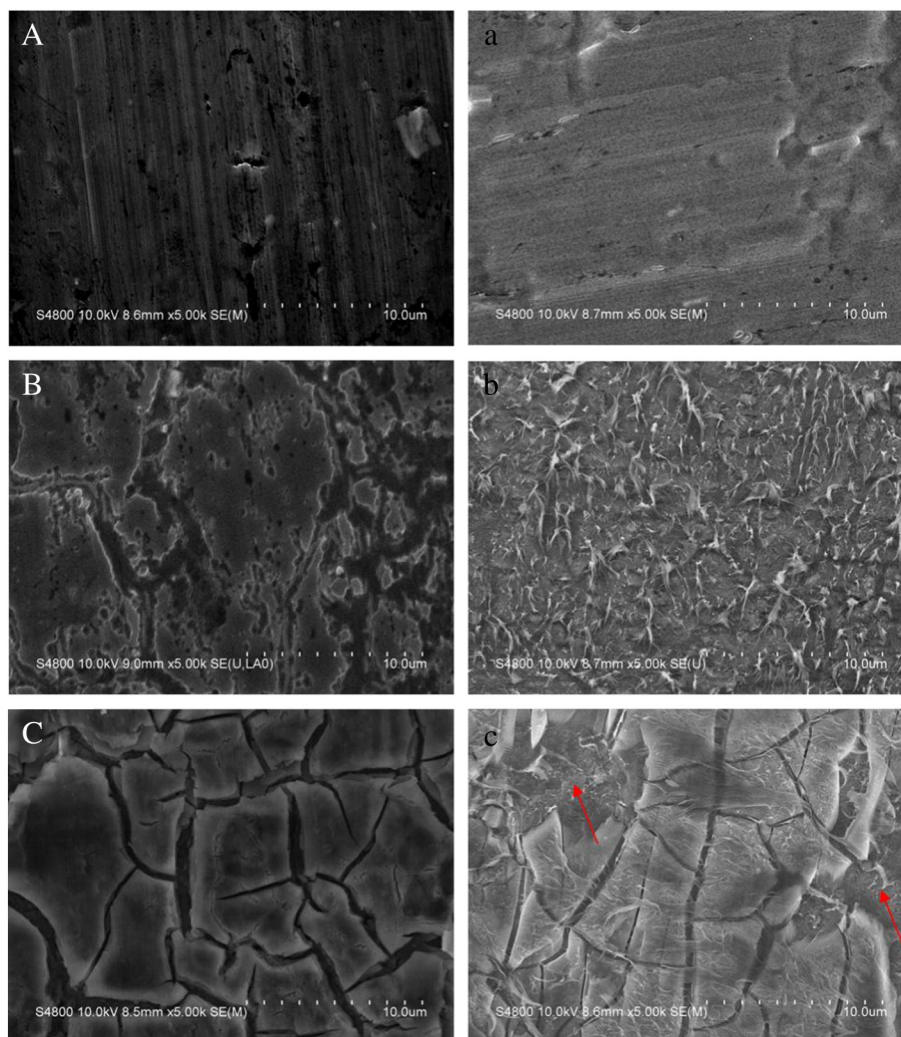


Fig. 2. Surface morphologies of Al foils before and after peeling with different pretreatment methods: (A, a) no treatment; (B, b) molybdate-based conversion; and (C, c) chromate-phosphate conversion.

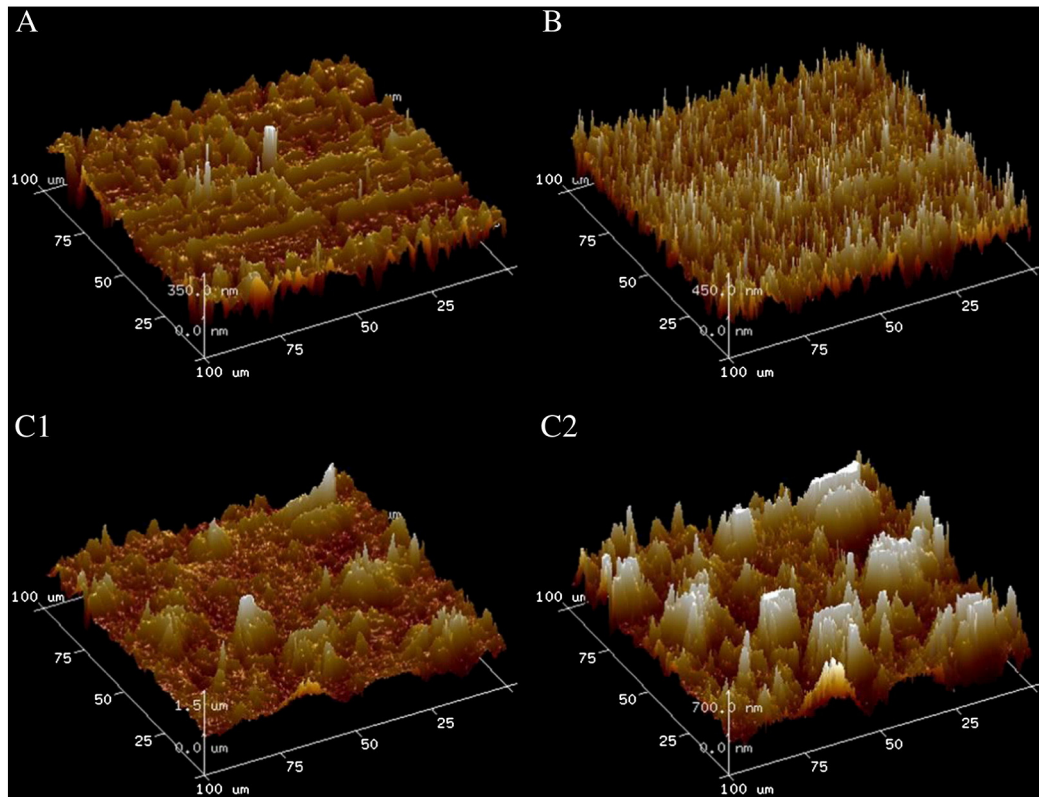


Fig. 3. AFM images of surface Al foils: (A) no treatment; (B) molybdate-based conversion; and (C1), (C2) chromate-phosphate conversion with a different Z coordinate.

ion (0.135 mol/L), phosphorate anion (0.211 mol/L), and fluorion (0.611 mol/L) at room temperature for 4 min. The samples were removed from the solution, rinsed in tap water, and finally dried in an oven for 1 h.

2.3. Preparation of adhesion sample

The Al/PP-g-GMA composites were prepared by hot pressing as follows. Al foil and PP-g-GMA were first placed in a mould composed of two pieces of polytetrafluoroethylene cloth. The mould was placed on a hot press, preheated for 15 min at 175 °C, and subjected to pressure for 40 s at 10 MPa. The composite sample was then air cooled at room temperature.

2.4. T-peel tests

The joint strength of the prepared Al/PP composites was measured by conducting a T-peel test according to the GB8808-88 method using a tensile tester CMT 2203 (SANS, China). The test was carried out at room temperature with a peeling speed of 300 mm/min. The peel strength was averaged over 5 samples.

Table 1

A summary of R_a values from AFM data.

	R_a (nm)
a	50.1
b	94.2
c	185.0

(a) No treatment, (b) molybdate-based conversion, and (c) chromate-phosphate conversion.

2.5. Surface analysis

2.5.1. SEM

The morphologies of the Al composite samples were examined using a scanning electron microscope S-4800 (Hitachi Limited, Japan) operated at 10 kV.

2.5.2. AFM

The surface roughness of the Al foils was examined using an atomic force microscope NanoScope IIIa MultiMode AFM (Veeco, America).

2.5.3. EDS

A QUANTAX 400-30 (Bruker AXS, Germany) energy-dispersive spectrometer was used to observe changes in the composition of the Al surface after treatment.

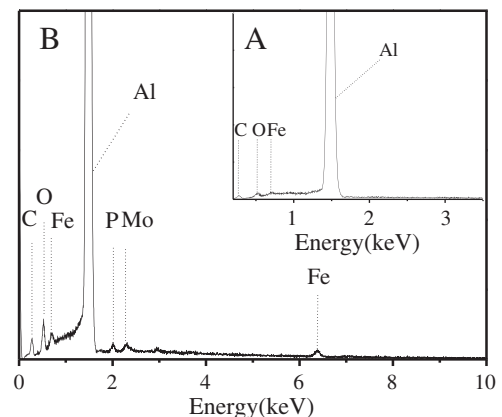


Fig. 4. EDS spectrum of (A) bare Al and (B) Al with molybdate-based conversion coating.

Table 2

Surface element compositions of bare Al and the treated Al foil with molybdate-based conversion coating obtained using EDS.

		Al	C	O	Fe	Mo	P
Bare foil	Wt.%	95.32	3.19	0.91	0.57		
	At.%	91.38	6.88	1.48	0.27		
Treated foil	Wt.%	88.14	5.25	3.46	1.24	1.57	0.34
	At.%	82.29	11.01	5.45	0.56	0.41	0.28

2.5.4. Contact-angle measurements

The surface wettabilities of bare Al and the chemical conversion coatings were determined by measuring the contact angle of deionized water on the surfaces using a contact angle goniometer JC2000D3 (Shanghai Zhongchen Digital Technic Apparatus, China). When the water drop made contact with the sample surface, timing was started and the image was collected, and the contact angle was measured at 15 s. All experiments were performed at 25 °C. The surface energy was calculated using Eq. (1), which was deduced by Kwork et al. [33].

$$\cos\theta + 1 = 2(\gamma_s/\gamma_L)^{\frac{1}{2}}e^{-\beta}(\gamma_L - \gamma_s)^2 \quad (1)$$

where θ is the contact angle, β is equal to $0.0001247 \text{ (mN/m)}^{-2}$, and γ_s and γ_L are the surface free energies of solid and liquid, respectively. The reported value was averaged over 5 samples.

3. Results and discussion

3.1. T-peeling tests

The T-peeling test is one of the most convenient methods of characterizing interface adhesion between metal and polymer. Considering that the T-peeling strength is influenced by surface properties, it can be used to evaluate the effects of pretreatment on Al foil. Fig. 1 shows the results of T-peeling strength measurements for untreated and treated Al foil/PP-g-GMA composite films.

The peeling strength between bare Al foil and PP-g-GMA is approximately $1 \pm 0.5 \text{ N/15 mm}$. The peeling strength between Al foil and PP-g-GMA significantly increases after surface treatment. Al foil treated with molybdate-based solution has a peeling strength of approximately $10 \pm 0.4 \text{ N/15 mm}$, and Al foil treated with chromate–phosphate conversion has a peeling strength of approximately $12 \pm 0.8 \text{ N/15 mm}$. The molybdate-based conversion treatment enhances the joining strength of Al and PP-g-GMA and its effect is approximate to chromate–phosphate conversion treatment.

3.2. Characteristics of the conversion coatings

To explore the effects of the surface properties of Al foil on the T-peeling strength of the composite, the surface morphology, chemical composition, and wettability of the foils were characterized by SEM, AFM, EDS, and contact-angle measurements.

3.2.1. Surface morphology analysis

Fig. 2(A)–(C) show the SEM images of untreated and treated Al surfaces. The morphology of untreated Al [Fig. 2(A)] has a smooth surface with the presence of some granules and holes. Regular scratches with rolling lines are also observed on the surface, which are possibly caused

by the fracture of brittle particles or embedding of debris. All these local defects likely result from the processing of Al. Fig. 2(B) shows that Al treated with molybdate-based solution has a ‘mud-cracked’ surface, wherein the coating forms small irregular block-like islands on Al. These cracks, which have also been observed by Juffs et al. [34] and Meng et al. [35], probably result from dehydration in air.

Al subjected to chromate–phosphate conversion shows a similar morphology, as shown in Fig. 2(C). The rough cracks found on the surface of Al can result in mechanical interlocking with the melted PP-g-GMA during hot pressing, which may partly explain the increase in peeling strength. The rough surface also has a greater contact area for bonding, which may lead to a proportionate increase in adhesion. A similar explanation has been given by Gent et al. [36], who investigated the peeling energy of rubber joined to smooth and grit-blasted steel. The grit-blasted steel exhibits a twofold to threefold increase in peeling energy, which is attributed to the increase in surface area.

Fig. 2(a)–(c) show the SEM images of the stripped surfaces of untreated and treated Al foils after T-peeling tests. Fig. 2(a) shows that no residual polymer (PP-g-GMA) remain on the bare Al surface after peeling because Al foil and PP-g-GMA have no interfacial chemical interaction, and only weak physical interactions exist between them. The obtained results agreed with the low peeling strength of bare Al/PP-g-GMA composite films.

Fig. 2(b) shows the SEM images of the Al surface after peeling PP-g-GMA film from Al foil treated with molybdate-based conversion treatment. Al foil is completely covered with residual polymers, and the stretched film can be clearly observed. These results indicated that PP-g-GMA endured plastic deformation, which can be explained by the effect of mechanical interlocking between the two rough surfaces. Chen et al. [4] drew a similar conclusion.

Fig. 2(c) shows that chromate–phosphate treated Al exhibits a different fracture surface. A ‘mud-cracked’ surface and some stretched polymer films can be found on the surface, and some conversion coatings appear to be exfoliated from the Al surface, as indicated by the arrow. This finding may be explained by the very high peeling strength caused by chromate–phosphate conversion, which locally exceeded the adhesive strength of the conversion coating on the Al surface. Consequently, peeling of the local conversion coating occurred during the T-peeling test.

Fig. 3 shows the AFM images of untreated and treated Al surfaces. It can be seen that the bare Al foil has a smooth surface with a Z coordinate 350 nm shown in Fig. 3(a). The Al foil treated with molybdate-based conversion treatment has a rougher surface, its Z coordinate is 450 nm, and there are many needle-like structures on the surface as shown in Fig. 3(b). Fig. 3(c1) and (c2) show the Al foil treated by chromate–phosphate conversion treatment, and they are located in a same place just with a different Z coordinate. It can be seen that the Al foil treated by chromate–phosphate conversion treatment has the roughest surface with a Z coordinate of 1.5 μm and 700 nm, respectively. Table 1 shows a summary of R_a values from AFM data, it can be concluded that the order of the surface roughness is $R_a(c) > R_a(b) > R_a(a)$, which corresponds to the AFM images and it also corresponds to the adhesion results.

3.2.2. Chemical composition analysis

Fig. 4 shows the EDS results of bare Al (Fig. 4a) and treated Al with molybdate-based conversion solution (Fig. 4b). Table 2 shows the surface element compositions of them determined from the EDS data.



Fig. 5. Spreading of a water droplet on the Al foil surface with different pretreatments: (A) no treatment, (B) molybdate-based conversion, and (C) chromate–phosphate conversion.

Table 3

Contact angle and surface energy between water and Al with different treatments.

	a	b	c
Contact angle (°)	77.9 ± 1.8	38.8 ± 2.2	15.5 ± 3.3
Surface energy (mN/m)	36.79 ± 1.44	60.02 ± 1.33	70.29 ± 1.77

(a) Untreated, (b) molybdate-based conversion treatment, and (c) chromate–phosphate conversion treatment.

Al foil with molybdate-based conversion coatings is mainly composed of the elements Al, Mo, C, O, P, and Fe (Fig. 4b). The presence of Mo and P, which is not observed in Fig. 4a, suggested that the treatment produced a new conversion coating that contained phosphate and molybdate. Changes in surface composition can affect joining strength. Koh et al. [38] used argon ion irradiation to pretreat PTFE both in the presence and absence of oxygen. High-resolution XPS spectra show chemical changes on the surface. The authors attribute the enhanced adhesion to the improved wettability and chemical reactivity of the surface combined with mechanical interlocking to the increasingly rough surface.

The Al surface displays a high polarity resulting from the presence of –OH groups which allows the formation of strong chemical interactions between GMA groups and –OH groups [37]. We can find that the content of O increased from 0.19 wt.% to 3.46 wt.%, so the –OH content is increased. We can deduce that there are more chemical interactions formed when the PP-g-GMA is hot pressed to the treated Al surface which induced a higher joining strength.

3.2.3. Contact-angle measurement

Fig. 5 presents the contact angle of Al foil treated with different methods. Both molybdate-based and chromate–phosphate pretreatments induced significant changes in the contact angle of Al.

Table 3 gives the contact angle and surface energy between water and Al with different treatments. Untreated Al has a large contact angle of 77.9°, indicating low hydrophilicity. Al samples treated with molybdate-based solution and chromate–phosphate solution have contact angles of 38.8° and 15.5°, respectively, indicating good hydrophilicity. The relative magnitude of the surface energies of the three Al foils followed the order chromate–phosphate treated (c) > molybdate-treated (b) > bare Al (a), as calculated from contact angles. The roughness of the molybdate-treated and chromate–phosphate-treated samples contributed to their water wetting.

4. Conclusions

A hydrophilic molybdate-based conversion coating for Al that enhanced the peeling strength between Al with PP-g-GMA was designed and prepared. The molybdate-based solution increased the peeling strength to a similar extent as chromate–phosphate conversion treatment and was environment friendly. Results indicated that molybdate-based conversion treatment was a promising alternative to chromate–phosphate conversion treatment for improving Al/P composites.

Conflict of interest

None.

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References

- [1] D.H. Berry, J. Adhes. 81 (2005) 347–370.
- [2] F. Xia, S.A. Xu, Appl. Surf. Sci. 268 (2013) 337–342.
- [3] S. Brunner, Ph. Gasser, H. Simmler, K.G. Wakili, Surf. Coat. Technol. 200 (2006) 5908–5914.
- [4] M.-A. Chen, H.-Z. Li, X.-M. Zhang, Int. J. Adhes. Adhes. 27 (2007) 175–187.
- [5] J. Qiu, E. Sakai, L. Lei, Y. Takarada, S. Murakami, J. Mater. Process. Technol. 212 (2012) 2406–2412.
- [6] S.G. Prolongo, A. Ureña, Int. J. Adhes. Adhes. 29 (2009) 23–31.
- [7] A.L. Gasparin, C.H. Wanke, R.C.R. Nunes, E.K. Tentardini, C.A. Figueroa, I.J.R. Baumvol, R.V.B. Oliveira, Thin Solid Films 534 (2013) 356–362.
- [8] S.M. Song, K. Cho, C.E. Park, H.K. Yun, S.Y. Oh, J. Appl. Polym. Sci. 85 (2002) 2202–2210.
- [9] A. Ghosh, D.A. Schiraldi, J. Appl. Polym. Sci. 112 (2009) 1738–1744.
- [10] J.H. Ku, I.H. Jung, K.Y. Rhee, S.J. Park, Compos. B 45 (2013) 1282–1287.
- [11] G.W. Critchlow, D.M. Brewis, Int. J. Adhes. Adhes. 16 (1996) 255–275.
- [12] B.N. Zand, M. Mahdavian, Surf. Coat. Technol. 203 (2009) 1677–1681.
- [13] H. Chen, J. Wang, Q. Huo, Thin Solid Films 515 (2007) 7181–7189.
- [14] F.M. De Wit, Ö. Özkanat, J.M.C. Mol, H. Terryn, J.H.W. De Wit, Surf. Interface Anal. 42 (2010) 316–320.
- [15] A.K. Mishra, R. Balasubramaniam, Corros. Sci. 49 (2007) 1027–1044.
- [16] J. Zhao, L. Xia, A. Sehgal, D. Lu, R.L. McCreery, G.S. Frankel, Surf. Coat. Technol. 140 (2001) 51–57.
- [17] L. Xia, E. Akiyama, G. Frankel, R. McCreery, J. Electrochem. Soc. 147 (2000) 2556–2562.
- [18] J. Zhao, G. Frankel, R.L. McCreery, J. Electrochem. Soc. 145 (1998) 2258–2264.
- [19] M.W. Kendig, A.J. Davenport, H.S. Isaacs, Corros. Sci. 34 (1993) 41–49.
- [20] L. Zhu, F. Yang, N. Ding, Surf. Coat. Technol. 201 (2007) 7829–7834.
- [21] O. Lunder, C. Simensen, Y. Yu, K. Nisancioglu, Surf. Coat. Technol. 184 (2004) 278–290.
- [22] H. Allachi, F. Chaouket, K. Draoui, J. Alloys Compd. 491 (2010) 223–229.
- [23] W.-K. Chen, J.-L. Lee, C.-Y. Bai, K.-H. Hou, M.-D. Ger, J. Taiwan Inst. Chem. E. 43 (2012) 989–995.
- [24] Z. Yong, J. Zhu, C. Qiu, Y. Liu, Appl. Surf. Sci. 255 (2008) 1672–1680.
- [25] C.G. da Silva, I.C.P. Margarit-Mattos, O.R. Mattos, H. Perrot, B. Tribollet, V. Vivier, Corros. Sci. 51 (2009) 151–158.
- [26] V. Karpakam, K. Kamaraj, S. Sathiyarayanan, G. Venkatachari, S. Ramu, Electrochim. Acta 56 (2011) 2165–2173.
- [27] Y.K. Song, F. Mansfeld, Corros. Sci. 48 (2006) 154–164.
- [28] A.A.O. Magalhães, I.C.P. Margarit, O.R. Mattos, J. Electroanal. Chem. 572 (2004) 433–440.
- [29] G. Wang, M. Zhang, R. Wu, Appl. Surf. Sci. 258 (2012) 2648–2654.
- [30] A.S. Hamdy, A.M. Beccaria, P. Traverso, J. Appl. Electrochem. 35 (2005) 467–472.
- [31] B.A. Shaw, G.D. Davis, T.L. Fritz, K.A. Olver, J. Electrochem. Soc. 137 (1990) 359–360.
- [32] C.-S. Liang, Z.-F. Lv, Y.-L. Zhu, S.-A. Xu, H. Wang, Appl. Surf. Sci. 288 (2014) 497–502.
- [33] D.Y. Kwork, A.W. Neumann, Adv. Colloid Interface Sci. 81 (1999) 167–249.
- [34] L. Juffs, A.E. Hughes, S. Furman, P.J.K. Paterson, Corros. Sci. 44 (2002) 1755–1781.
- [35] Q. Meng, G.S. Frankel, Surf. Interface Anal. 36 (2004) 30–42.
- [36] A.N. Gent, S.-M. Lai, Rubber Chem. Technol. 68 (1995) 13–25.
- [37] J.C. Bolger, in: K.L. Mittal (Ed.), Adhesion Aspects of Polymeric Coatings, Plenum Press, 1983, p. p3.
- [38] S.K. Koh, S.C. Park, J.W. Seok, S.C. Choi, W.K. Choi, H.J. Jung, Mater. Res. Soc. Symp. Proc. 438 (1997) 523–528.